

Assignment 6 Report

Financial Engineering — Structured Products

Group 5

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Abstract

This report presents the results for Assignment 6. The first part prices the mid-market upfront $X\%$ of a 2-year certificate swap on the EURO STOXX 50 under a calibrated NIG model and compares it with two Black benchmarks (smile-adjusted and flat). We then extend the analysis to the 3-year version of the same product with double trigger and discuss the loss of the Black alternative. The second part prices a 10-year Bermudan Payer Swaption on the Euribor 3m curve, non-call 2, strike 5%, under a single-factor Hull-White model with $a = 0.11$ and $\sigma = 0.008$. We use a trinomial tree on the zero-mean OU state variable, verify the implementation against a market ZCB and against the Jamshidian decomposition of the longest European, and compare the Bermudan price with analytical Jamshidian-based bounds.

Case Study 1: Certificate Pricing

On 15 February 2008 Bank XX issues a 2-year certificate with notional $N = 100$ Mln EUR (Annex 1) and the hedging swap of Annex 2 with counterparty I.B. At the Start Date (19 February 2008) I.B. pays $X\% \cdot N$. Both legs receive or pay the structured coupon and Euribor 3m + 1.30% respectively, subject to the Early Redemption clause: if the Year-1 coupon resets at 6% (i.e. Stoxx50 below the Strike $K = 3200$ at the Reset Date), the swap cancels at the first payment.

The mid-market upfront equalises the two legs at the Start Date. We use the NIG (Normal Inverse Gaussian) parameters (σ, κ, η) calibrated on the EURO STOXX market surface, and the discount curve bootstrapped in Assignment 2. Interest rates and equity dynamics are assumed independent.

Pricing scheme

The certificate value depends on a single trigger event at the Year-1 Reset Date T_{reset} (two business days before the first payment date t_1). There are two mutually exclusive scenarios:

- **Trigger** ($S(T_{\text{reset}}) < K$). The first coupon resets at 6%, the Cumulative Coupon Accrual exceeds the 6% trigger and the swap cancels at t_1 . Bank XX pays Euribor + spread up to t_1 and receives the 6% coupon at t_1 .
- **No trigger** ($S(T_{\text{reset}}) \geq K$). The Year-1 coupon is 0, the swap survives, and the final 2% coupon is paid at t_2 . Bank XX pays Euribor + spread up to t_2 .

Denoting $p = \mathbb{Q}(S(T_{\text{reset}}) < K)$, the upfront is computed as:

$$X = p \cdot \left[(1 - B(0, t_1)) + s \text{BPV}_1 \right] + (1 - p) \cdot \left[(1 - B(0, t_2)) + s \text{BPV}_2 \right] - p B(0, t_1) \alpha_1 c_1 - (1 - p) B(0, t_2) \alpha_2 c_2,$$

where $s = 1.30\%$ is the floating spread, $c_1 = 6\%$ and $c_2 = 2\%$ are the potential coupons, α_i are the fixed-leg accrual factors (30/360 mod. following) and BPV_i are the floating-leg Basis Point Values (quarterly Euribor 3m, Act/360). The model enters only through the trigger probability p . All other quantities are model-independent.

a) NIG upfront

We use the NIG forward dynamics on F_t (under the T_{reset} -forward measure) for estimating the trigger probability via Monte Carlo:

$$F_t = F_0 \exp\left(\sigma\sqrt{t}\sqrt{G}Z - (0.5 + \eta)\sigma^2 t G - \ln \mathcal{L}(\eta; t)\right), \quad G \sim \text{IG}(1, t/\kappa), \quad Z \sim \mathcal{N}(0, 1),$$

The term $\ln \mathcal{L}(\eta; t)$ is the Laplace correction, which ensures the martingale property $\mathbb{E}[F_t] = F_0$. We use the same horizon $t = T_{\text{reset}}$ of the calibration to ensure consistency between calibrated parameters and simulated distribution. We simulate 10^7 independent realizations of $F_{T_{\text{reset}}}$ and we estimate the trigger probability in the following way:

$$P_{\text{trigger}} = \frac{1}{N_{\text{sim}}} \sum_{i=1}^{N_{\text{sim}}} \mathbb{I}_{\{F_{T_{\text{reset}}}^{(i)} < K\}}$$

We use a fixed seed for reproducibility and we compute the upfront by using the above formula.

Quantity	Value
F_0 (forward at T_{reset})	3,692.16
NIG parameters (σ, κ, η)	(0.135, 1.00, 3.00)
Trigger probability p	0.5125
Mid-market upfront X^{NIG}	5.8951%
Upfront on $N = 100$ Mln	5,895,127 EUR

Table 1: Point (a) – NIG upfront on the 2-year certificate.

b) Model selection

An alternative pricing route uses Black with a *smile-adjusted digital*, where the trigger probability is recovered as

$$p = 1 - \frac{\widehat{\text{Dig}}_{\text{call}}}{B(0, T_{\text{reset}})}, \quad \widehat{\text{Dig}}_{\text{call}}(K) = -\frac{\partial C(K, \sigma(K))}{\partial K} \approx \frac{C(K) - C(K + \varepsilon)}{\varepsilon},$$

with both $\sigma(K)$ and $\sigma(K + \varepsilon)$ interpolated on the same spline of the market vol surface. This formulation includes the local smile slope $-\text{Vega}(K) \partial \sigma / \partial K$, which dominates the gap with respect to the flat-Black quote. The result, reported in Table 2, is within 4 bps of the NIG value, confirming that the smile is the main driver in this regime.

Model	X^*	$X^* \cdot N$ (EUR)
NIG (point a)	5.8951%	5,895,127
Black, smile-adjusted (point b)	5.8608%	5,860,788
Difference (b – a)	–3.4 bp	–34,339

Table 2: Point (b) – Alternative pricing under Black with smile correction.

The two models agree closely because the certificate depends on a single equity observation at $T_{\text{reset}} \approx 1$ y. At one date the only model-dependent quantity is the marginal density at T_{reset} . Both NIG and smile-adjusted Black calibrate to the same market call prices, so their implied marginals at K are very similar. The residual ~ 4 bp gap is the effect of the local second-order curvature (smile convexity) captured by NIG but not by a first-order slope correction.

c) Three-year extension: is the Black alternative still available?

In the 3-year version with the Year-1 payoff repeated at Year 2 and Early Redemption at the end of Year 1 and Year 2, two equity observations enter the pricing: $S(T_1)$ and $S(T_2)$. The trigger sequence and the corresponding cash flows depend on the *joint* distribution of $(S(T_1), S(T_2))$:

- $S(T_1) < K$: trigger at Year 1, swap cancels, 6% paid at t_1 .
- $S(T_1) \geq K$ and $S(T_2) < K$: trigger at Year 2, swap cancels, 6% paid at t_2 .
- $S(T_1) \geq K$ and $S(T_2) \geq K$: survive to maturity, 2% paid at t_3 .

A Black model with a flat or smile-adjusted volatility at strike K only fixes the marginal density of S at one date. There is no canonical way to extend it to a two-time joint density without bringing in extra unverified assumptions (e.g. constant volatility forward dynamics with a chosen correlation), and the result is sensitive to those assumptions. The NIG Lévy process, by contrast, defines a consistent joint law over multiple time horizons through its independent stationary increments. **The Black alternative of point (b) is therefore *not* a satisfactory substitute for the NIG model in the 3-year case.**

d) Three-year upfront under NIG with deterministic rates

We extend the certificate pricing to the 3-year case keeping the NIG parameters of point (a), the deterministic rate curve from the Assignment 2 bootstrap and the dividend yield of point (a). Following the hint, we simulate the spot under the risk-neutral measure as

$$S_{T_i} = \frac{S_0}{B(0, T_i)} e^{-q T_i} e^{f_{T_i}},$$

where $B(0, T)$ is the curve discount factor and f_t is the NIG log-price (NMVM with $\alpha = 1/2$). Exploiting the Lévy property, the log-price increment between the two observation dates is split into two independent variables $\Delta f_1 \sim \text{NIG}(\sigma, \kappa, \eta; \Delta t_1)$ and $\Delta f_2 \sim \text{NIG}(\sigma, \kappa, \eta; \Delta t_2)$, with $\Delta t_1 = T_1^{\text{reset}}$ and $\Delta t_2 = T_2^{\text{reset}} - T_1^{\text{reset}}$. The deterministic part is factorised through the curve discount factors,

$$\text{grow}_1 = \frac{e^{-q \Delta t_1}}{B(0, T_1^{\text{reset}})}, \quad \text{grow}_2 = \frac{e^{-q \Delta t_2} B(0, T_1^{\text{reset}})}{B(0, T_2^{\text{reset}})},$$

so that the chained spot reads $S_{T_1} = S_0 \text{grow}_1 e^{\Delta f_1}$ and $S_{T_2} = S_{T_1} \text{grow}_2 e^{\Delta f_2}$.

On each path at most one trigger fires: $\{S_{T_1} < K\}$ calls the certificate at T_1 , otherwise $\{S_{T_2} < K\}$ calls it at T_2 ; if neither barrier is breached the path survives to T_3 and receives the reduced 2% coupon. The structured leg (Investment Bank) and the floating leg (Bank XX) are discounted with the bootstrapped curve, keeping in mind that the floater stops at the trigger date: the identity $1 - B(0, T_{\text{end}})$ and the BPV that prices the spread therefore use a scenario-dependent T_{end} (T_1 , T_2 or T_3). The mid-market upfront is computed on $N_{\text{sim}} = 10^7$ paths as $X = (\text{PV}^{\text{floater}} - \text{PV}^{\text{coupon}})/N$.

Quantity	Value
$\mathbb{Q}(\text{trigger at } T_1)$	0.2455
$\mathbb{Q}(\text{trigger at } T_2 \mid \text{no trigger at } T_1)$	0.2014
$\mathbb{Q}(\text{survival to } T_3)$	0.6025
Mid-market upfront $X^{\text{NIG}, 3Y}$	8.2274%
Upfront on $N = 100$ Mln	8,227,381 EUR

Table 3: Point (d) – 3-year NIG upfront.

The 3-year upfront is larger than the 2-year one, and the intuition is not the extra trigger chance but rather the longer floating leg on the survivors. Two effects compete. On one side, the 3-year contract gives a second opportunity to get the high 6% coupon: the conditional probability $\mathbb{Q}(\text{trigger at } T_2 \mid \text{no trigger at } T_1) = 0.2014$ lifts the total mass of paths paying the 6% from $\approx 24.6\%$ in the 2Y to $\approx 39.7\%$ in the 3Y, which is an outflow for IB. On the other side, the 60.25% of paths that survive to T_3 keep paying Euribor+spread to IB for an extra year, while IB only owes the reduced 2% consolation coupon at maturity. The floating leg PV scales with the contract horizon through $1 - B(0, T_{\text{end}})$, and this longer floating inflow on the majority of paths dominates the cost of the additional digital. The net effect is favourable to IB, so the upfront it pays to Bank XX at inception has to grow in order to bring the contract back to zero value.

e) Error of the naive Black model

The naive Black approach prices each digital with a single implied volatility, $\sigma(K)$, read from the market surface at the strike of interest, with no smile adjustment. For the 2-year contract the trigger probability reads

$$p^{\text{Black flat}} = \Phi(-d_2), \quad d_2 = \frac{\ln(F_0/K) - \frac{1}{2}\sigma(K)^2 T}{\sigma(K)\sqrt{T}},$$

i.e. a lognormal density at T_{reset} with a single fixed vol parameter. For the 3-year contract the same single σ is used in the chained MC of points (c)/(d): both log-return increments $\Delta f_1, \Delta f_2$ are drawn from $\mathcal{N}(-\frac{1}{2}\sigma^2 \Delta t, \sigma^2 \Delta t)$, so the joint law of (S_{T_1}, S_{T_2}) is bivariate lognormal. In both cases the smile slope is ignored.

Maturity	Model	X^*	$X^* \cdot N$ (EUR)
2Y	NIG (reference, point a)	5.8951%	5,895,127
	Black, flat σ (point e)	5.2597%	5,259,680
	Error (Black – NIG)	–63.5 bp	–635,447
3Y	NIG (reference, point d)	8.2274%	8,227,381
	Black, flat σ (point e)	7.0834%	7,083,355
	Error (Black – NIG)	–114.4 bp	–1,144,027

Table 4: Point (e) – Mispricing of the naive Black model at the two maturities.

In both cases the flat Black model underprices the upfront. On the 2-year contract the error is –63.5 bp on the notional ($\sim 635,000$ EUR). With the negative skew of the EURO STOXX surface the smile-correction is negative, so the true (smile-aware) trigger probability is *lower* than the one returned by flat Black at $\sigma(K)$. The naive model therefore *overestimates* the trigger probability, which has two reinforcing effects on the upfront: IB looks like it will pay the high 6%

coupon more often, and the floating leg paid by Bank XX terminates more often at T_1 instead of T_2 , shrinking its PV. Both push the balancing upfront down with respect to the NIG reference.

On the 3-year contract the error nearly doubles, to -114.4 bp on the notional (~ 1.14 Mln EUR). The same smile-blindness mechanism is at play, but it now compounds across two consecutive digital observations: each reset date is mispriced, and the chained joint law of (S_{T_1}, S_{T_2}) inherits the mismatch on both legs. The error therefore grows roughly proportionally with the number of digital fixings, not with the contract horizon.

Case Study 2: Bermudan Swaption under Hull-White

We price a 10-year Bermudan Payer Swaption with annual exercises starting from Year 2 (non-call 2), strike $K = 5\%$, written on a 10-year annual fixed-versus- Euribor-3m swap that matures on 19 February 2018. The single-curve framework uses the discount factors bootstrapped from Euribor 3m on 15 February 2008. Under the one-factor Hull-White model

$$dr(t) = (\theta(t) - ar(t))dt + \sigma dW(t), \quad a = 0.11, \quad \sigma = 0.008,$$

we work on the zero-mean OU state variable $x(t) = r(t) - \alpha(t)$ and build a trinomial tree on x .

a) Trinomial-tree pricing

The tree is piecewise-uniform in time: only the contractual dates (start, the eight exercise dates and maturity) are guaranteed nodes, and within each contractual segment the sub-grid is strictly uniform with $\sim \text{stepsPerYear}$ steps per year. This guarantees that Δt is constant inside each segment, which is the assumption of all Hull-White closed-form formulas (probabilities, $\hat{\mu}$, $\hat{\sigma}$, $\hat{\sigma}^*$, stochastic discounts). Δt varies slightly across segments, but each backward-induction step uses its own local Δt_i , so there is no inconsistency.

We also considered the alternative of using a single constant Δt over the whole horizon and snapping each exercise date to the nearest tree node (which introduces a small timing error of at most $\Delta t/2$). We chose the piecewise-uniform grid instead, so that exercise dates fall exactly on nodes and the timing of the early-exercise decision is exact. In both approaches what matters for the tree to recombine is that Δx is constant, not Δt .

The spatial grid has constant step $\Delta x = \hat{\sigma}_{\text{ref}}\sqrt{3}$ for tree recombination, with $\hat{\sigma}_{\text{ref}}$ based on the average Δt , and is truncated at $\pm l_{\text{max}} = \lceil (1 - \sqrt{2/3})/\hat{\mu}_{\text{ref}} \rceil$ where the branching switches to upward / downward shapes. The one-step stochastic discount follows the lecture-notes decomposition

$$D(t_i, t_{i+1}) = B(t_i, t_{i+1}; x_i) \cdot \exp \left\{ -\frac{1}{2}(\hat{\sigma}^*)^2 - \frac{\hat{\sigma}^*}{\hat{\sigma}} [\Delta x_{i+1} + \hat{\mu} x_i] \right\},$$

where $B(t_i, t_{i+1}; x_i)$ is the analytical HW bond at the current node. At each exercise date the option value is the maximum between the continuation value and the intrinsic Payer Swap value computed via the analytical HW bond formula for all residual swap payment dates.

The Bermudan price is stable from 12 steps per year onwards, in agreement with the expected $\mathcal{O}(\Delta t)$ convergence of the trinomial-tree scheme:

Steps per year	Total tree nodes	Bermudan price
1	11	0.020850
4	41	0.021101
12	121	0.021108
52	521	0.021118
365	3655	0.021119

Table 5: Bermudan price convergence with the number of steps per year.

The fair price of the Bermudan Payer Swaption on a unit notional is

$$V^{\text{Berm}} = 0.02112.$$

b) Sanity checks on the tree

We verify the implementation with two independent checks.

Check 1 — 10-year ZCB calibration. Setting the terminal payoff to 1 at maturity and running pure backward induction without any exercise must return the market discount factor $B^M(0, T)$. This isolates the tree calibration to the initial curve. Results in Table 6.

Steps per year	Total nodes	Tree ZCB price
1	11	0.64702
4	41	0.64733
12	121	0.64740
52	522	0.64743
365	3655	0.64744
Market $B^M(0, 10\text{y})$		0.64417
Error at 365 steps		$3.27 \cdot 10^{-3}$

Table 6: Tree convergence of the 10-year ZCB versus the market discount factor.

The tree converges to a value approximately $3 \cdot 10^{-3}$ above the target market value, and the error does not vanish as $\Delta t \rightarrow 0$. In our opinion this is due to a structural feature of the decomposition: while $\mathbb{E}_{\mathbb{Q}}[D(t_i, t_{i+1}) \mid \mathcal{F}_{t_i}] = B(t_i, t_{i+1}; x_i)$ is exact in continuous time, the trinomial branching only matches the first two conditional moments of Δx_{i+1} and not the full Gaussian distribution. The accumulated mismatch in the joint law of $(x_1, \dots, x_N)$ produces a residual calibration bias that does not converge to zero with Δt .

Check 2 — 9-year European swaption vs. Jamshidian. Restricting the exercise set to a single date at Year 9 turns the Bermudan into a European swaption on the residual one-period swap. We pick the longest available exercise (Year 9) on purpose, so that the tree has to perform a deep backward induction over the full horizon and the check stresses the tree implementation as much as possible. The closed-form benchmark is the Jamshidian decomposition, which prices the European swaption as a portfolio of put options on Zero Coupon Bonds.

Steps per year	Total nodes	Tree European price
1	11	0.004408
4	41	0.004403
12	121	0.004397
52	522	0.004386
365	3655	0.004387
Jamshidian closed form		0.004728
Error at 365 steps		$3.41 \cdot 10^{-4}$

Table 7: 9-year European swaption: tree convergence vs. Jamshidian formula.

The tree converges to a value $\sim 3.4 \cdot 10^{-4}$ below the Jamshidian benchmark. The pattern (the same direction and order of magnitude as the ZCB check, after accounting for the fact that

the swaption depends on the entire term structure of HW bonds at the exercise date) confirms that the residual is a discretization artefact of the chosen stochastic-discount decomposition and not a bug in the swaption-payoff implementation.

c) Bounds for the Bermudan price

By no-arbitrage, the Bermudan price V^{Berm} must lie between a lower and an upper bound. We derive three analytical quantities via the Jamshidian decomposition.

Lower bound. The Bermudan can always replicate any single-exercise European policy, so it is worth at least as much as the most valuable European swaption within the exercise schedule:

$$LB = \max_{i=1,\dots,n} V_i^{\text{Eur}},$$

where V_i^{Eur} is the European swaption price with exercise at T_i on the residual swap $[T_i, T_N]$, computed by Jamshidian decomposition.

Loose upper bound. If each exercise opportunity could be exploited independently (one separate European swaption per date), the total value would be at least as large as the Bermudan, which allows only one exercise:

$$UB_{\text{eur}} = \sum_{i=1}^n V_i^{\text{Eur}}.$$

This bound is valid but *loose*: each V_i^{Eur} is priced on the full residual swap from T_i to T_N , so the same future cash flows are counted multiple times across different i .

Tight upper bound. A tighter bound is obtained by decomposing the swap payoff *period by period*. Using $(\sum_j x_j)^+ \leq \sum_j (x_j)^+$ on each period's cash flow and the sub-martingale property of $(F_j(\cdot) - K)^+$ under the T_j -forward measure (which makes the optimal time to lock in F_j equal to T_{j-1} , the last available exercise before the reset), one obtains

$$V^{\text{Berm}} \leq UB_{\text{cap}} = \sum_{j=3}^{10} \text{Caplet}_j,$$

where Caplet_j is the standard European caplet for the period $[T_{j-1}, T_j]$ with strike K . There are eight such caplets, one per Bermudan exercise date. Each is priced analytically under Hull-White via the put-on-ZCB equivalence:

$$\text{Caplet}_j = (1 + K \alpha_j) \cdot B(0, T_{j-1}) [K_j^* \Phi(-d_2) - F_j \Phi(-d_1)],$$

with $K_j^* = 1/(1 + K \alpha_j)$, $F_j = B(0, T_j)/B(0, T_{j-1})$, and $\sigma_{p,j} = (\sigma/a)(1 - e^{-a\Delta_j})\sqrt{(1 - e^{-2aT_{j-1}})/(2a)}$. Each future cash flow is counted *exactly once*, making UB_{cap} much tighter than UB_{eur} .

Steps/yr	Bermudan	LB	UB_{cap} (tight)	UB_{eur} (loose)	Check
1	0.020850	0.01486	0.02446	0.08898	OK
4	0.021101	0.01486	0.02446	0.08898	OK
12	0.021108	0.01486	0.02446	0.08898	OK
52	0.021118	0.01486	0.02446	0.08898	OK
365	0.021119	0.01486	0.02446	0.08898	OK

Table 8: Bermudan price against the three analytical bounds. The OK/FAIL check uses the tight interval $[LB, UB_{\text{cap}}]$.

Results. The Bermudan price is consistent with both bounds at every precision level. Three observations follow from the numbers.

1. **Gap above the lower bound.** $V^{\text{Berm}} \approx 0.0211$ is about 42% above $LB = 0.0149$. The difference captures the value of having eight exercise opportunities rather than only one: the holder benefits from the ability to wait and observe the short rate before deciding when to enter the swap.
2. **Gap below the tight upper bound.** V^{Berm} sits about 14% below $UB_{\text{cap}} = 0.0245$. This gap is small. It reflects the fact that the Bermudan must commit to *all* residual cash flows at once when it exercises, whereas the cap strip lets each period be independently exercised: the Bermudan gives up the option to selectively avoid negative-carry periods.
3. **Loose upper bound.** $UB_{\text{eur}} = 0.0890$ is about $4\times$ larger than UB_{cap} . The excess arises because each V_i^{Eur} is priced on the full residual swap, so the later periods (e.g. the period $[T_9, T_{10}]$) are counted eight times — once for each exercise date. The tight bound counts every period exactly once, making it far more informative.

Overall, the interval $[0.0149, 0.0245]$ given by the tight bounds is informative: it shows that the Bermudan price is well-determined up to a 15% relative band, and the tree price at 365 steps falls inside this band at all precision levels tested.

Problems and Criticalities

The two sanity checks agree on the source of the residual error in the trinomial tree. With the stochastic-discount decomposition we follow from the theory seen in class, the tree carries an intrinsic, Δt -independent calibration bias of order 10^{-3} on the long end (10y) of the curve. The bias propagates into derivative prices: the 9-year European swaption is offpriced by $\sim 7\%$ against its Jamshidian benchmark. For the Bermudan we have no closed-form benchmark, but the bias is likely of similar relative size. The Jamshidian-based interval $[0.0149, 0.0890]$ accommodates this, so the sanity check is non-binding, but the user of the tree should be aware of the limit when reporting absolute prices.

A side observation from an early mistake. In our first implementation of the tree we made a subtle but instructive error: instead of discounting one step with the model-implied stochastic bond $B(t_i, t_{i+1}; x_i)$, we mistakenly used the deterministic market forward $B^M(0, t_{i+1})/B^M(0, t_i)$. Conceptually these two objects are very different: the first is random and known only at time t_i , the second is deterministic and known at time 0. Yet, in practice, this “wrong” decomposition produced surprisingly good numbers: the ZCB calibration error collapsed to machine precision, the 9-year European swaption matched the Jamshidian closed form to within 10^{-7} , and the Bermudan price was very close to the textbook result. We then realised the discrepancy with the course material and reverted to the orthodox scheme that uses the model-implied one-step bond, as specified in the slides – which is the version reported in the previous paragraphs and the one we take as our reference price.

For completeness, we report below the two sets of results at 365 steps per year. The interesting takeaway is that the “orthodox” decomposition still carries a small residual bias on the ZCB and on the European swaption, because the trinomial branching only matches the first two conditional moments of Δx_{i+1} , while the accidental variant anchored the discount directly to the initial market curve and therefore wiped out the calibration error by construction. It is a useful reminder that a scheme which is theoretically less rigorous can still be numerically self-consistent on this kind of quantity, and we kept it visible here as an honest record of the path we walked.

Quantity	Orthodox scheme	Early variant
ZCB calibration error (10y)	$3.27 \cdot 10^{-3}$	$1.26 \cdot 10^{-13}$
9y European swaption error	$3.41 \cdot 10^{-4}$	$8.52 \cdot 10^{-7}$
Bermudan price (365 steps)	0.02112	0.02142

Table 9: Comparison between the orthodox one-step discount used in the body of the report and the early variant we accidentally implemented first.

Functions Implemented

- `[parameters, sigma, kappa, eta] = compute_parameters(data, dates, discounts, strike, spread, volatilities, strikes)`
- `upfront = compute_upfront(parameters, sigma, kappa, eta, N_sim, alpha, first_coupon, second_coupon, spread, notional, method)`
- `upfront = compute_upfront_3y(parameters, sigma, kappa, eta, N_sim, alpha, first_coupon, second_coupon, spread, notional)`
- `BPV = BasisPointValueFloating(reference_date, maturity_date, dates, discounts)`
- `residuals = calibration_residuals(p, x, vol_mkt, t, B, F0, rate, alpha, N, dx)`
- `price = Price_FFT_NMV(x_grid, sigma, kappa, eta, t, N, dx, B, F0, alpha)`
- `val = integrand_NMV_Lewis(z, t, sigma, kappa, eta, alpha)`
- `dg_price = digital_price_smile(parameters)`
- `[dg_put, dg_call] = price_digital_Black(parameters, sigma)`
- `[prices, table] = run_hw_pricing_swaption(a, sigma, K, startDate, exerciseDates, swapPaymentDates, precision_levels, dates, discounts, instrument_type)`
- `price = price_bermudan(a, sigma, K, dt_vector, grid_dates, l_max, dx, start_date, curve_dates, curve_discounts, swap_payment_dates, idx_exercise)`
- `price = price_zcb_tree(a, sigma, dt_vector, grid_dates, l_max, dx, start_date, curve_dates, curve_discounts)`
- `price = price_jamshidian(a, sigma, K, t_exercise, swap_pay_dates, start_date, curve_dates, curve_discounts)`
- `[LB, UB_eur, UB_cap] = compute_bermudan_bounds(a, sigma, K, exercise_dates, swap_pay_dates, start_date, curve_dates, curve_discounts)`
- `[time_grid, grid_dates, dt_vector, idx_exercise] = build_hw_time_grid(startDate, maturityDate, stepsPerYear, exerciseDates)`
- `payoff = calc_swap_payoff(x_nodes, t_exercise, K, a, sigma, T_payments, B_t0_t_exercise, B_t0_t_payments, start_date)`
- `ZCB = compute_hw_zcb(x_nodes, t_curr, T_future, a, sigma, B0_t_curr, B0_T_future, start_date)`
- `[Du, Dm, Dd] = compute_stochastic_discount(x_nodes, B_t_t1, dx, mu_hat, a, sigma, dt)`
- `validation_table = verify_swaption_bounds(precision_levels, prices, LB, UB)`
- `caplet_price = compute_hw_caplet(a, sigma, K, T_reset, T_payment, start_date, curve_dates, curve_discounts)`